Segmentation QC Guide

Thank you for using the spatial segmentation wizard! Here we've compiled a reference for interpreting the metrics produced by the spatial segmentation QC pipeline. Good segmentation balances transcript capture, real-cell recovery, elimination of outlier sizes, and faithful morphological representation of diverse cell types. In our studies we have found Baysor to be the leading tool that maximizes these qualities but encourage researchers to test the many tools we offer as results may vary.

Goals of Good Segmentation

- **Maximize transcript capture** — assign as many transcripts to cells as possible (% Transcripts Captured).
- **Recover real cells** — detect biological cells accurately without over- or under-segmenting (Cells Detected).
- **Eliminate outliers** — reduce very small or very large segments that inflate gene-expression overlap and confuse downstream annotation.
- **Respect morphology** — represent irregular shapes (fibroblasts, neurons, vessels) faithfully, not just circular blobs.

Expression QC Metrics (Report Page 1)

Cells Detected

Total number of cell segments identified by each method. More cells is not always better: over-segmentation splits real cells into fragments (inflating counts), while under-segmentation merges neighbors. Compare against expected tissue density and DAPI / morphology images to judge whether the count is reasonable. This is easiest done in Xenium explorer or other software for your modality.

% Transcripts Captured

Fraction of all detected transcripts (from the raw Xenium transcript table) that fall inside any cell boundary. Computed as:

```
% captured = 100 × sum(count_matrix) / total_transcripts
```

Higher values mean fewer transcripts are lost to the extracellular space. Typical values range from 50–85% depending on tissue type and segmentation aggressiveness.

% Cells Without Nucleus

Percentage of cell segments that do not spatially overlap a detected nucleus. Computed by projecting each Xenium nucleus centroid into cell polygons via a spatial join. Cells with no matched nucleus may represent cytoplasmic fragments, debris, or misaligned boundaries. Lower is generally better, though some biological contexts (e.g. enucleated red blood cells, platelets) naturally lack nuclei.

Median Counts / Genes per Cell

Median total transcript counts and median number of unique genes detected per cell. These reflect segmentation sensitivity: methods that draw tighter boundaries may report higher medians (capturing denser transcript clusters) but at the cost of missing peripheral transcripts. The violin/box plots show the full per-cell distribution — watch for long tails of near-zero cells (empty segments) or extremely high counts (merged segments).

Counts vs. Genes Scatter

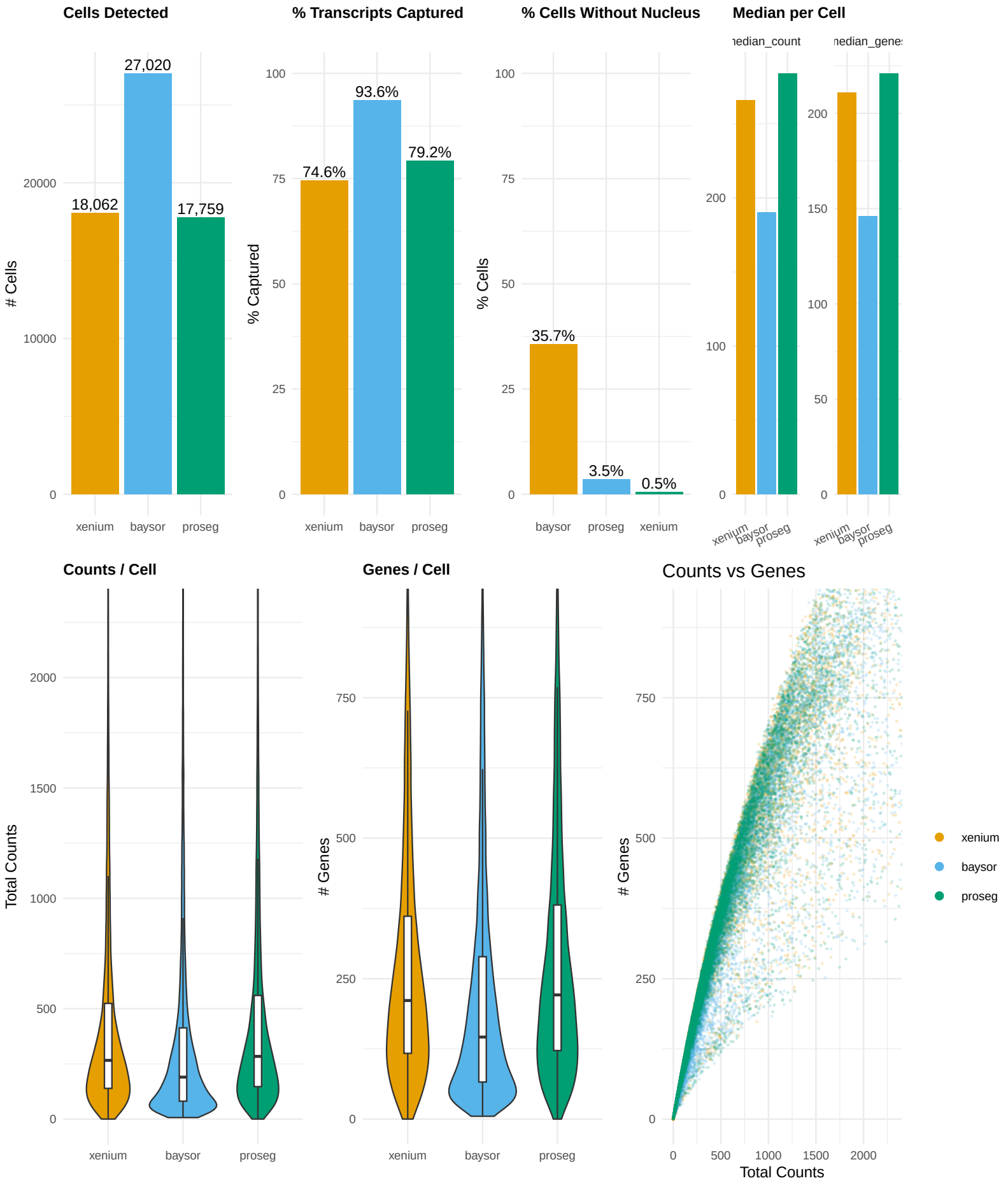
Each point is one cell; x-axis is total counts, y-axis is unique genes. A healthy segmentation shows a tight, monotonically-increasing cloud. Dispersion / horizontal streaks along the graph (many counts but few genes) suggest contamination or doublets. A low-count cluster near the origin indicates empty or debris segments.

Whats in this .pdf? What are our future directions?

Below you will find the outputs for these metrics for the methods that you ran. While helpful nothing is better than the human eye and we encourage researchers to look through many examples of their data to gain ideas of where segmentation is working well and not working as good.

*** Future versions of this pipeline will also assess these metrics by cell-type using a trained gradient boosting annotation tool from a reference dataset…. Stay tuned ***

Segmentation QC Report − JDMXeniumRawData_Slide1ROI6_TestData



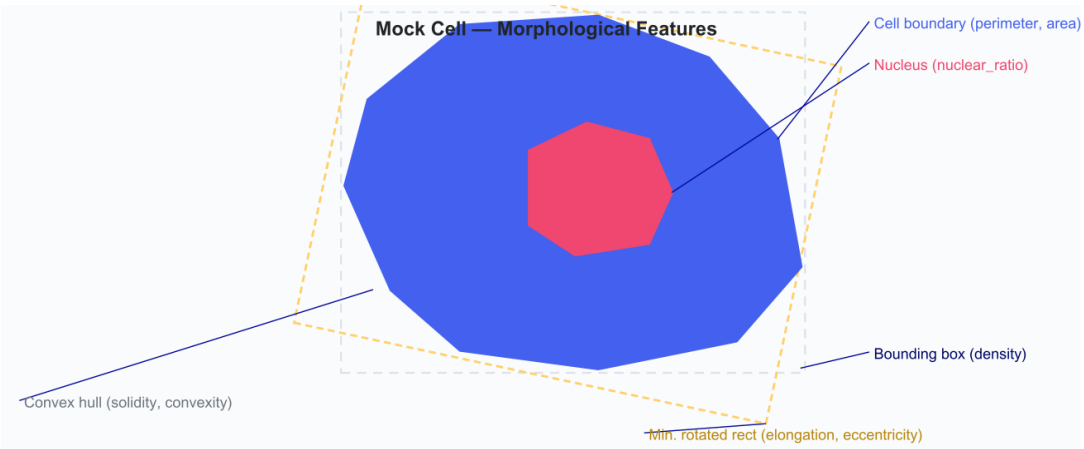
Morphological Metrics (Report Page 2)

Each metric below is computed per cell from the polygon boundary geometry using Shapely. Distributions are displayed as violin + box plots across methods. The table lists every formula, its expected range, and what deviations indicate. Many of these metrics are taken directly from the R package CellSPA and we recommend reading their API for more information.

Metric	Formula	Range	Interpretation
cell_area	`polygon.area`	Tissue-dependent	Extreme outliers suggest over/under-segmentation. Compare against known cell diameters for your tissue type.
elongation	`major_axis / minor_axis`	≥1 (1 = circular; >3 elongated)	Neurons and fibroblasts are naturally elongated. Round cell types (lymphocytes) should cluster near 1.
circularity	`4π · area / perimeter²`	0–1 (1 = perfect circle)	Low values indicate spiky or irregular boundaries, possibly from noisy / pixelated detection.
compactness	`sqrt(4π · area) / perimeter`	0–1 (1 = perfect circle)	Like circularity but less sensitive to perimeter noise. Prefer this when comparing methods with different boundary smoothness.
eccentricity	`sqrt(1 − (minor/major)²)`	0–1 (0 = circle, ~1 = line)	Bounded complement to elongation. Useful for flagging extremely stretched segments.
solidity	`area / convex_hull_area`	0–1 (1 = fully convex)	Low solidity means deep boundary indentations — can indicate star like cell morphology or segmentation artifacts.
convexity	`hull_perimeter / perimeter`	0–1 (1 = fully convex)	Low values flag wavy or fractal-like boundaries, often a sign of noisy pixelated segmentation.
density	`area / (bbox_w × bbox_h)`	0–1 (1 = fills bbox)	Very low density suggests large empty regions in the bounding box — typical of elongated or curved shapes.
nuclear_ratio	`nucleus_area / cell_area`	0–1 (typical 0.1–0.5)	Very high (>0.8) = boundary too tight (just the nucleus) typical in lymphocytes. Very low (<0.05) = over-expanded boundary. NA if no nucleus matched.

Visual Reference — Mock Cell Mask

The diagram below illustrates how morphological features are derived from a single cell's polygon boundary. Each geometric construct (convex hull, bounding box, minimum rotated rectangle, nucleus overlay) maps directly to one or more metrics in the table above.



Blue = cell boundary | Red = nucleus | Gray dashed = convex hull | Amber dotted = min. rotated rectangle | Gray dotted = bounding box

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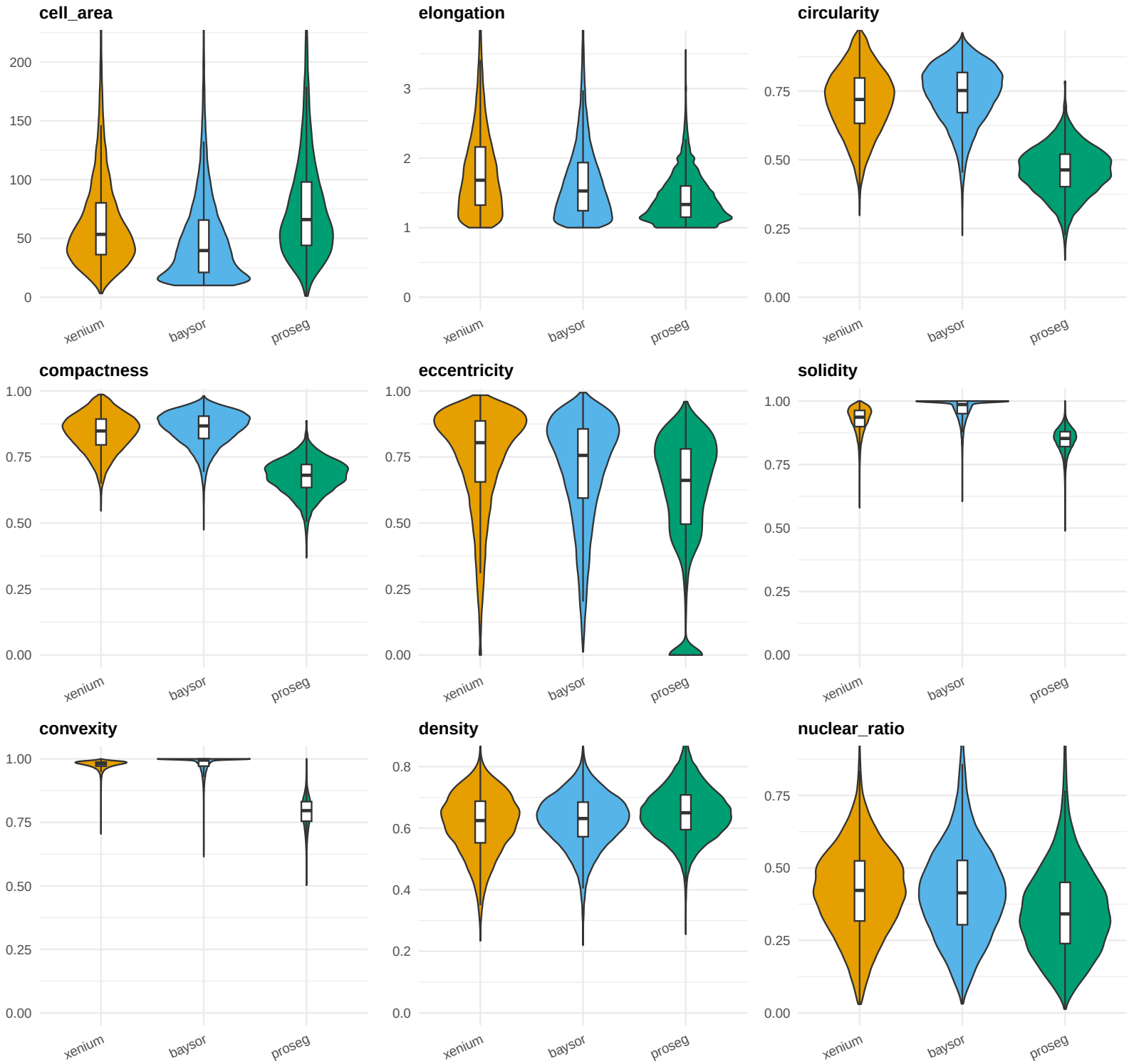
Practical Tips for Choosing a Segmentation

1. Nothing beats the human eye!
2. No single metric is sufficient — compare the full profile across methods.
3. If two methods have similar transcript capture, prefer the one with tighter morphological distributions (lower variance in circularity, solidity, etc.).
4. High '% Cells Without Nucleus' combined with many small cells suggests over-segmentation of cytoplasmic fragments.
5. For tissues with diverse cell types (e.g. brain), expect wider elongation and circularity distributions — this is biological signal, not an artifact.
6. Use the counts-vs-genes scatter to spot systematic doublets (upper-right outliers) and empty segments (lower-left cluster near the origin).

Morphological Metrics

Per−cell distributions computed from segmentation boundary geometry